MATH 201 Your name: ____________________

Learning target DFa, version 1

Correctly interpret the meaning of the derivative in context. Assign the correct units to the value of the derivative.

A company manufactures rope, and the total cost of producing r feet of rope is $C(r)$ dollars.

1. Suppose $C(2000) = 800$. What are the units on the 2000? What are the units on the 800?

2. Write a sentence explaining what $C(2000) = 800$ means in the story.

3. Suppose $C'(2000) = 0.35$. What are the units on the 2000? What are the units on the 0.35?

4. Write a sentence explaining what $C'(2000) = 0.35$ means in the story.
   Don't say "per," and don't say "rate."

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Learning targets DF1 and DFa, version 2

Correctly interpret the meaning of the derivative in context. Assign the correct units to the value of the derivative.

Arapaho Glacier is a mountain glacier in Roosevelt National Forest, west of Boulder, CO. The following table[1] gives the surface area, $A(t)$, in square meters, of Arapaho Glacier in the year t.

t	1900	1960	1973	1999
$A(t)$	338,282	250,764	225,000	162,027

1. Compute an approximation for $A'(1960)$, and **include units** for this number.

2. Write a sentence explaining what the number means about how the area of the glacier is changing. **Don't say "per," and don't say "rate."**

3. Compute an approximation for $A'(1999)$, and **include units** for this number. Do you think your approximation is too high or too low? Why?

4. How does $A'(1960)$ compare to $A'(1999)$? Is that good or bad?

[1] Haugen, B., Scambos, T., Pfeffer, T., & Anderson, R. (2010). Twentieth-century changes in the thickness and extent of Arapaho Glacier, Front Range, Colorado. *Arctic, Antarctic, and Alpine Research, 42*(2), 198-209.

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Learning target DFa, version 3

Correctly interpret the meaning of the derivative in context. Assign the correct units to the value of the derivative.

Let $f(t)$ be the number of centimeters of rain that have fallen by time t hours after midnight.

(a) Suppose $f(3) = 1.5$. What are the units on the 3? What are the units on the 1.5?

(b) Write a sentence explaining what $f(3) = 1.5$ means in the story.

(c) Suppose $f'(3) = 0.2$. What are the units on the 3? What are the units on the 0.2?

(d) Write a sentence explaining what $f'(3) = 0.2$ means in the story.
Don't say "per" and don't say "rate".

(e) Would the statement $f'(5) = -0.7$ make sense? Why or why not?

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Learning target DFa, version 4

Correctly interpret the meaning of the derivative in context. Assign the correct units to the value of the derivative.

In order to prepare for the upcoming ski season, a local resort is making snow on some of its lower trails. The snow depth, in inches, at time t is $D(t)$, where t is the number of days we are into December (so, for instance, $t = 9$ is December 9).

(a) Suppose $D(15) = 28$. What are the units on the 15? What are the units on the 28?

(b) Write a sentence explaining what $D(15) = 28$ means in the story.

(c) Suppose $D'(15) = 1.4$. What are the units on the 15? What are the units on the 1.4?

(d) Write a sentence explaining what $D'(15) = 1.4$ means in the story.
Don't say "per" and don't say "rate".